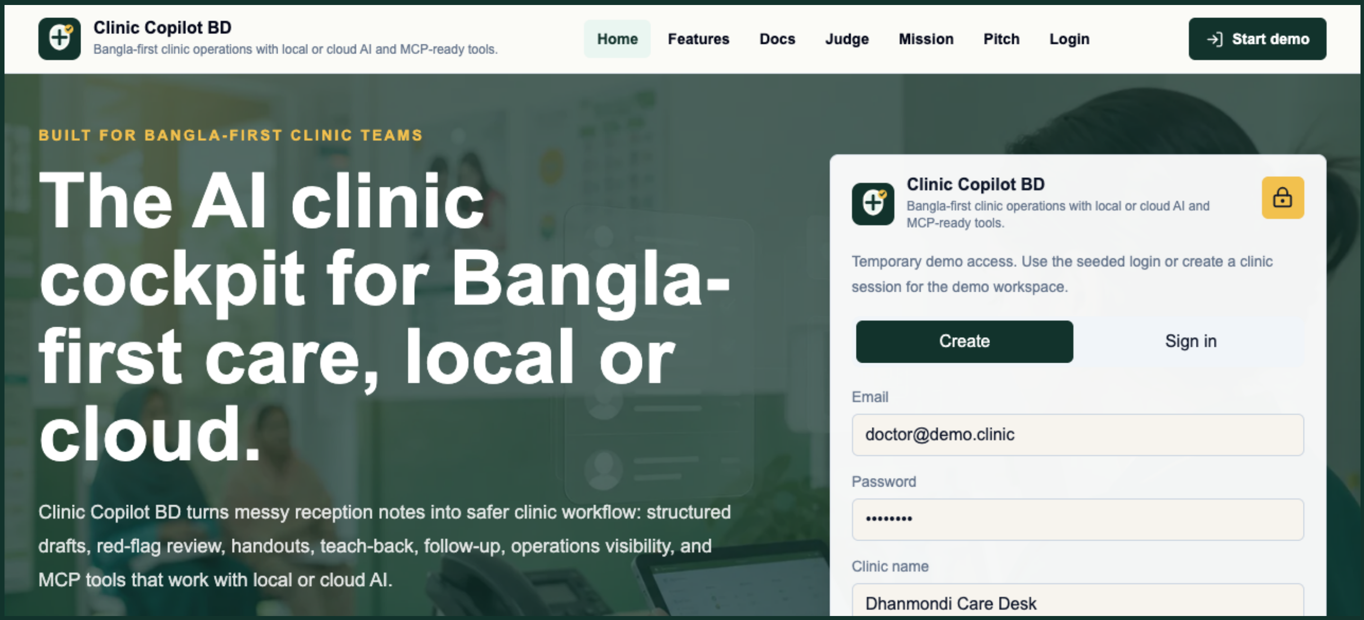
Clinic Copilot BD

Health Worker Assistant for Bangladesh Frontline Care

AI decision support for small hospitals, overloaded clinics, field health workers, and low-resource primary care teams

Local or cloud AI | MCP-ready | Human review | Bangla-first workflow



Hackathon segment	Health Worker Assistant: an AI decision-support tool for field health workers to improve frontline healthcare delivery.
Audience	Investors, hospital owners, clinic operators, hackathon judges, healthcare researchers, NGO/public-health partners, and marketing teams.
Team	Huntriox. Lead: Touhidul Alam Seyam. Contact: seyamalam41@gmail.com.
Report status	Live prototype evidence, fresh screenshots, MCP architecture, synthetic-data demo, production hardening plan.
Date	June 11, 2026.

Contents

1 Executive Summary

Clinic Copilot BD is a Bangla-first AI workflow assistant for smaller hospitals, private clinics, community clinics, and frontline health workers. It helps a real care team move from a messy patient story to a safer next action without pretending to replace a clinician.

Product thesis. The best health-worker AI for Bangladesh is not the most medically aggressive chatbot. It is the one that reduces workload, catches missing checks, keeps outputs draft-only, works with local or cloud models, and makes human review auditable.

1.1 What the product does

1. Captures intake in Bangla, English, or mixed language.
2. Converts rough notes into structured draft support.
3. Flags missing vitals, allergy gaps, pregnancy/child/chest-pain risk, and return-warning blockers.
4. Produces clinician-reviewable handoff, print packet, patient education, and follow-up tasks.
5. Keeps an audit trail and requires human review for clinical or patient-facing outputs.
6. Exposes the same safe workflows through MCP so external agents and local AI hosts can call bounded tools.

1.2 Primary wedge

The first commercial wedge is the smaller hospital or overloaded clinic: facilities with heavy patient flow, thin staffing, limited IT capacity, and urgent need for intake, queue, documentation, patient education, and follow-up support.

1.3 Expansion path

The same system expands into larger hospitals through outpatient throughput, emergency intake support, nursing handoff, discharge education, quality audit, training, and research evaluation.

2 Team Huntriox

Figure 1: Team Huntriox and Clinic Copilot BD project identity.

Team name: Huntriox. The team combines software engineering, AI product development, and computer engineering perspectives from Bangladesh. The project is led by Touhidul Alam Seyam. Primary contact: seyamalam41@gmail.com.

Touhidul Alam Seyam Team Lead
Department of Computer Engineering, BGC Trust University Bangladesh
Also Software Engineer at Agentic AI Institute. Leads product architecture, AI workflow, MCP integration, safety model, and demo execution.

Abtahee Kabir Team Member
Department of Computer Engineering, CUET
Supports engineering, implementation review, technical validation, and hackathon delivery.

Chandnin Barua Jowthi Team Member
Department of Computer Engineering, BGC Trust University Bangladesh
Supports product research, user perspective, healthcare workflow framing, and presentation readiness.

3 Fresh Product Screenshots

The following screenshots were captured from the running app for this PDF report using the project browser QA flow and a focused asset set. They show the main surfaces a judge, buyer, researcher, or investor should inspect.

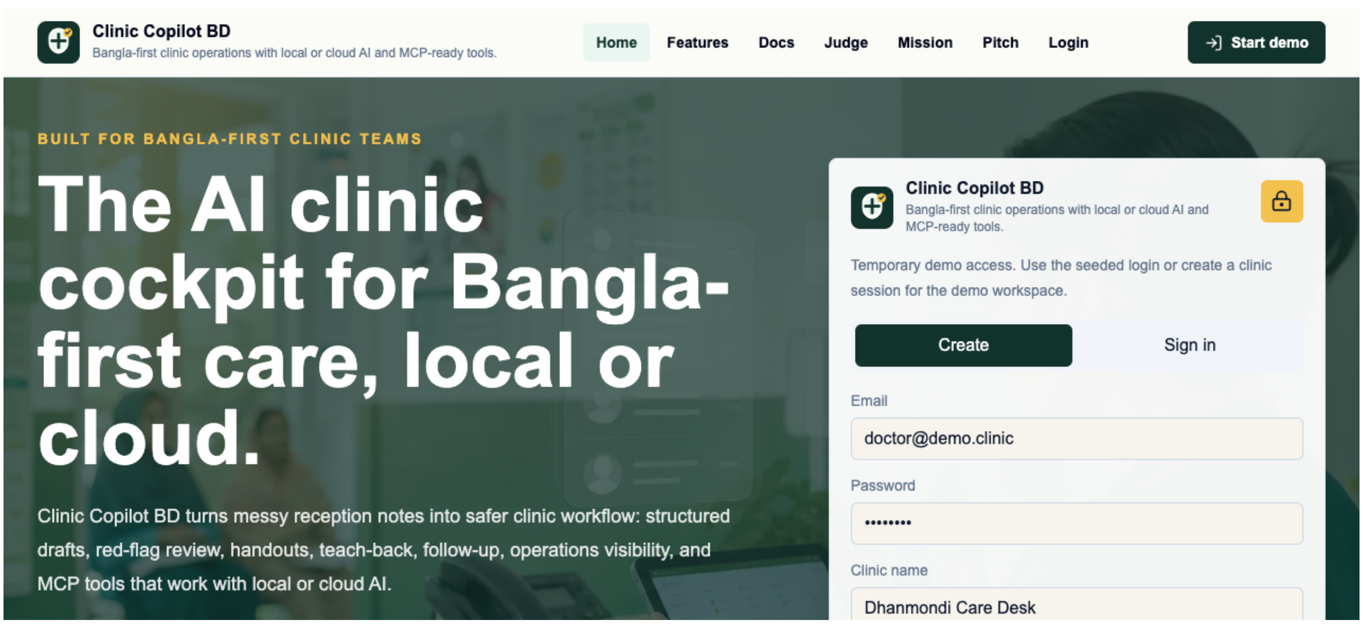


Figure 2: Public home and product positioning
The first impression frames Clinic Copilot BD as a practical health worker assistant with local/cloud AI and MCP-ready tools.

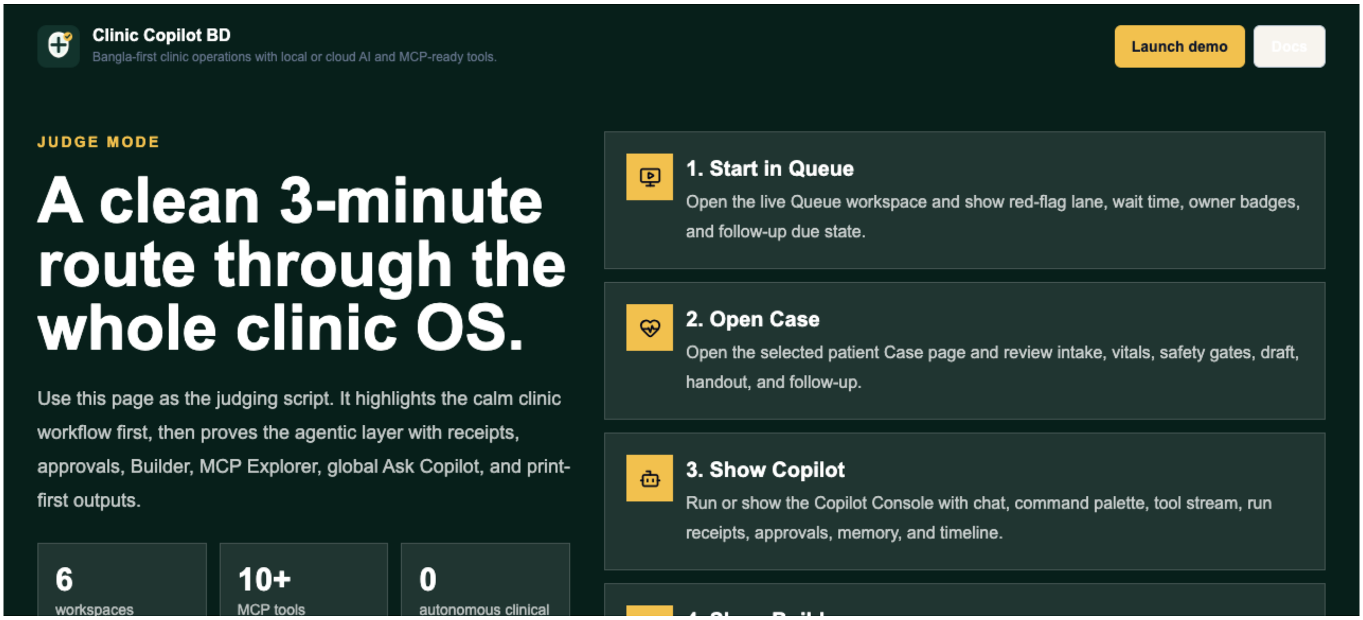


Figure 3: Judge mode
A short hackathon evaluation path for reviewers: open queue, show case workflow, show safety, show Copilot, prove MCP.

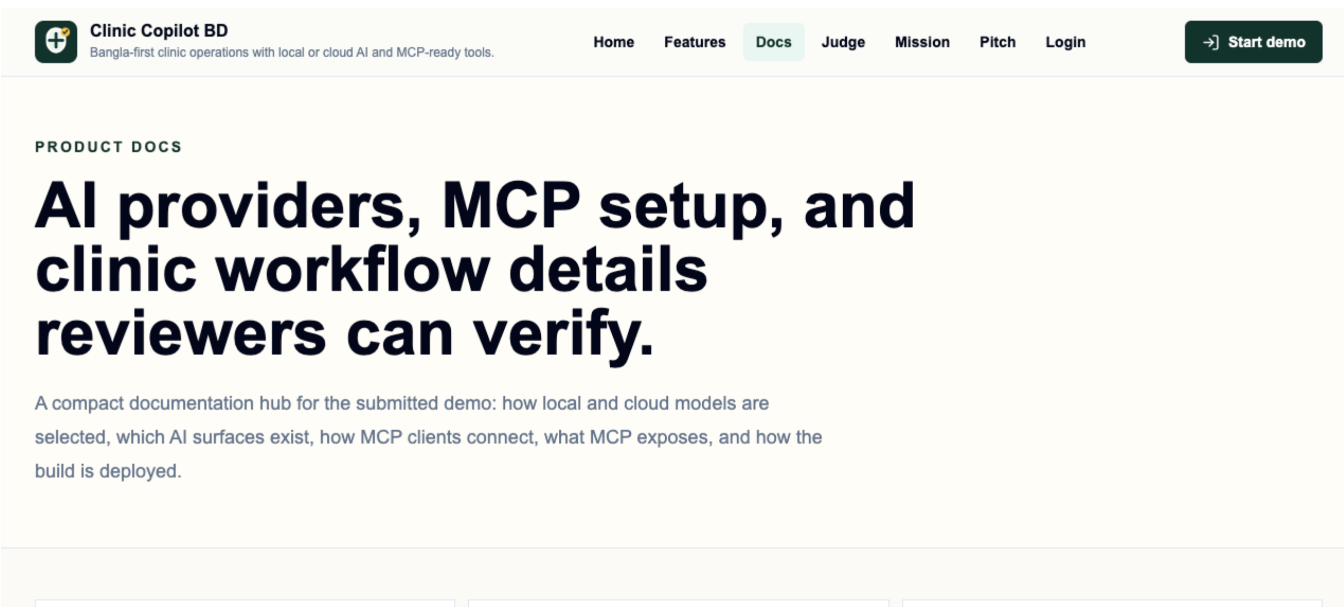


Figure 4: Documentation and MCP proof path
The docs give technical reviewers a reproducible path through local AI, cloud AI, MCP, and the demo workflow.

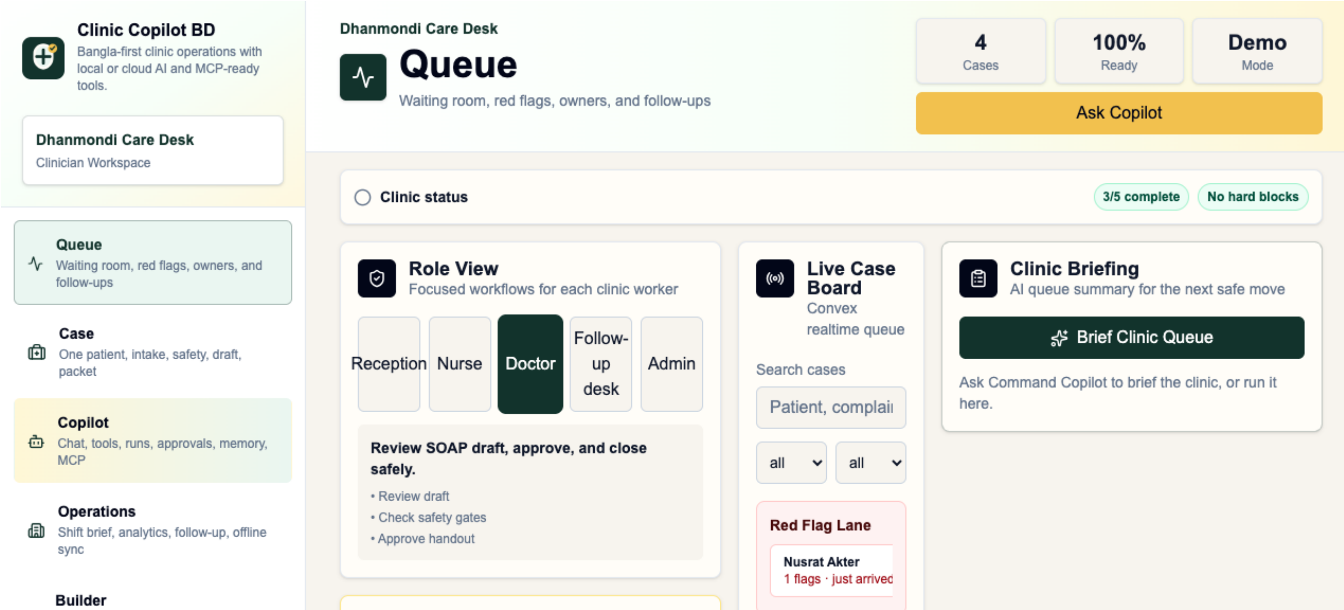


Figure 5: Live queue workspace
Queue pressure, red-flag lane, owners, and follow-up status are visible for overloaded front desks.

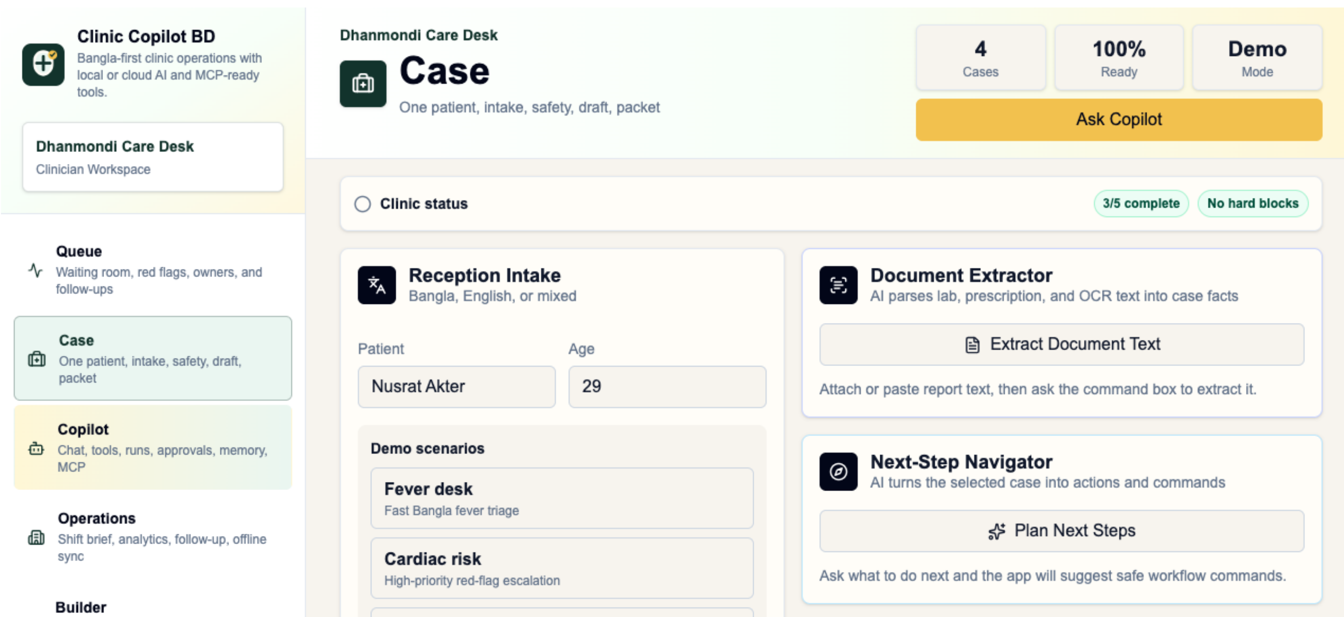


Figure 6: Case workspace and clinical safety gates
Intake, missing checks, safety gates, draft support, handout, medicine safety, and approval guard are grouped around the patient case.

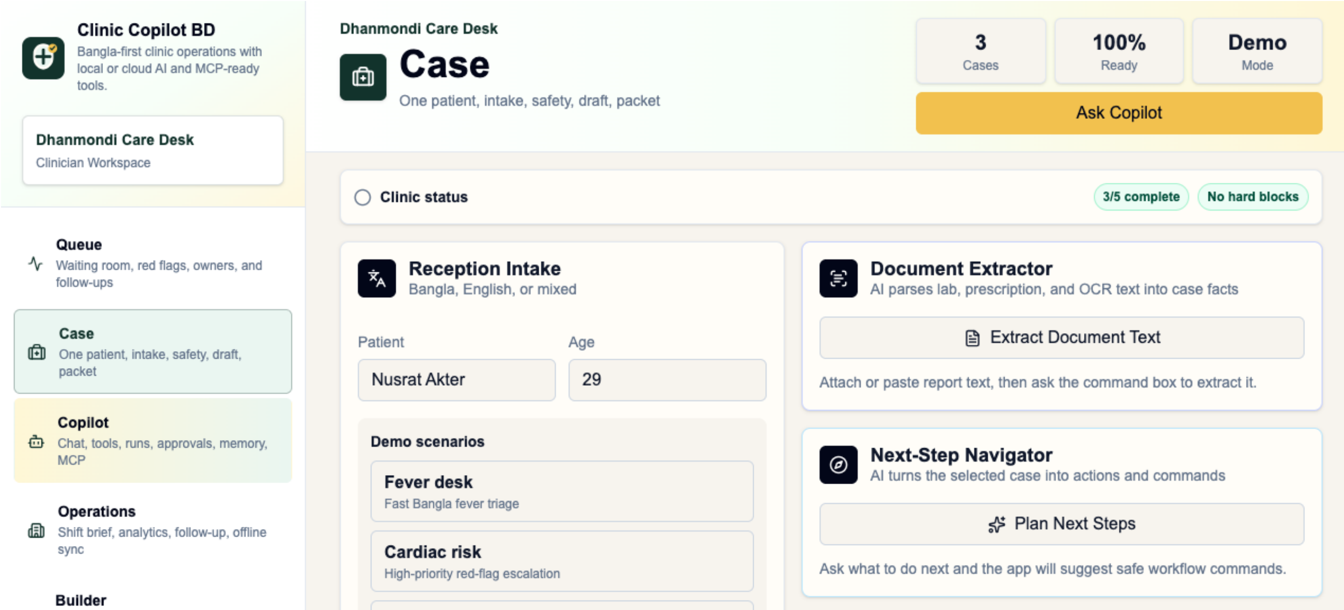


Figure 7: Generated draft and safety workflow
The app exercises AI generation and safety checks while keeping outputs draft-only.

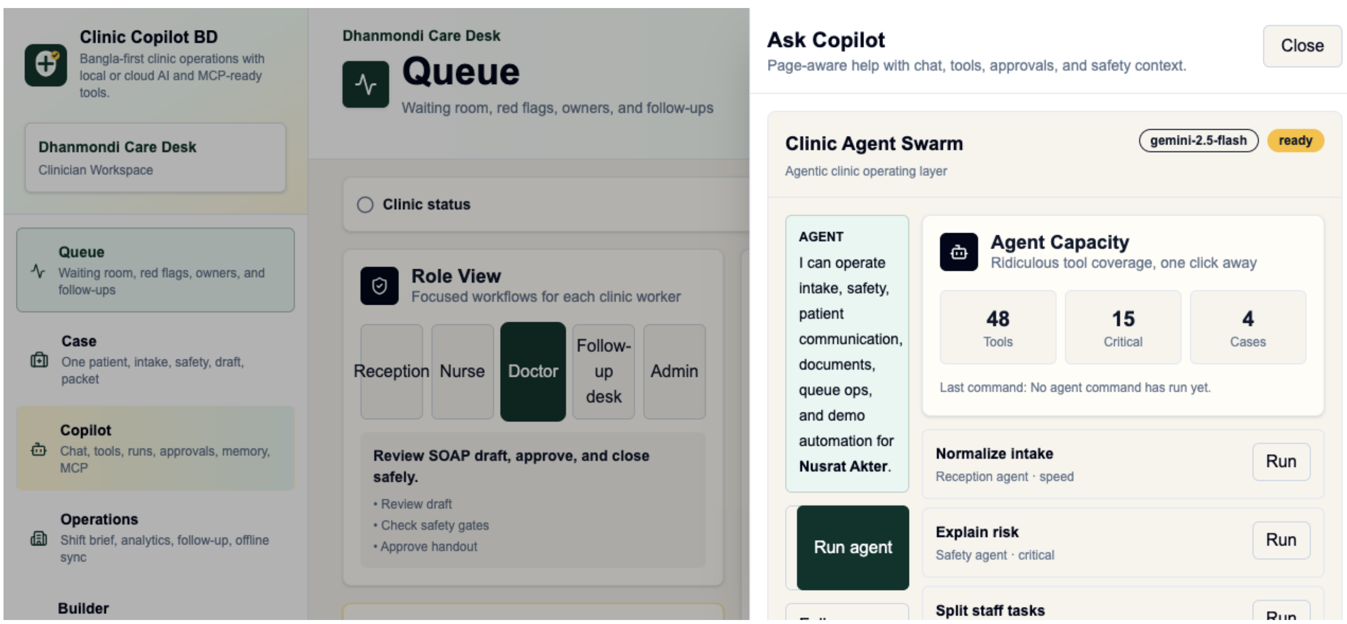


Figure 8: Copilot, run receipts, and approvals
Global Ask Copilot connects to AI run receipts and an approvals inbox for auditable human review.

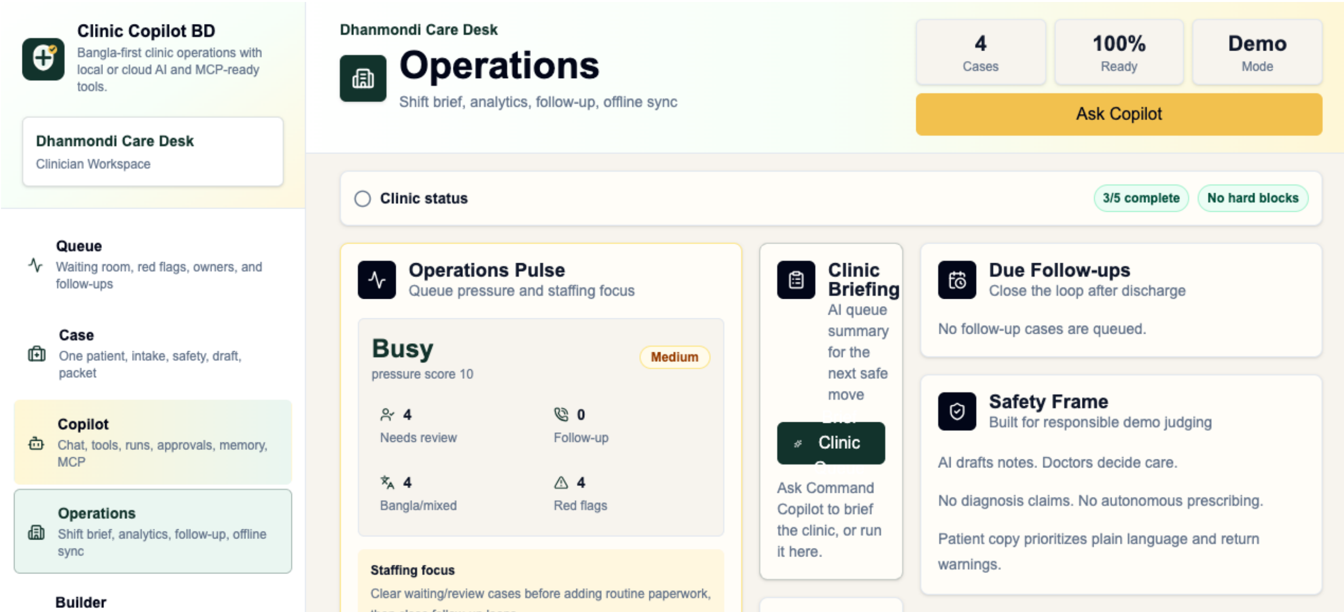


Figure 9: Operations pulse
Supervisors can see queue, follow-up, and shift pressure instead of only one patient at a time.

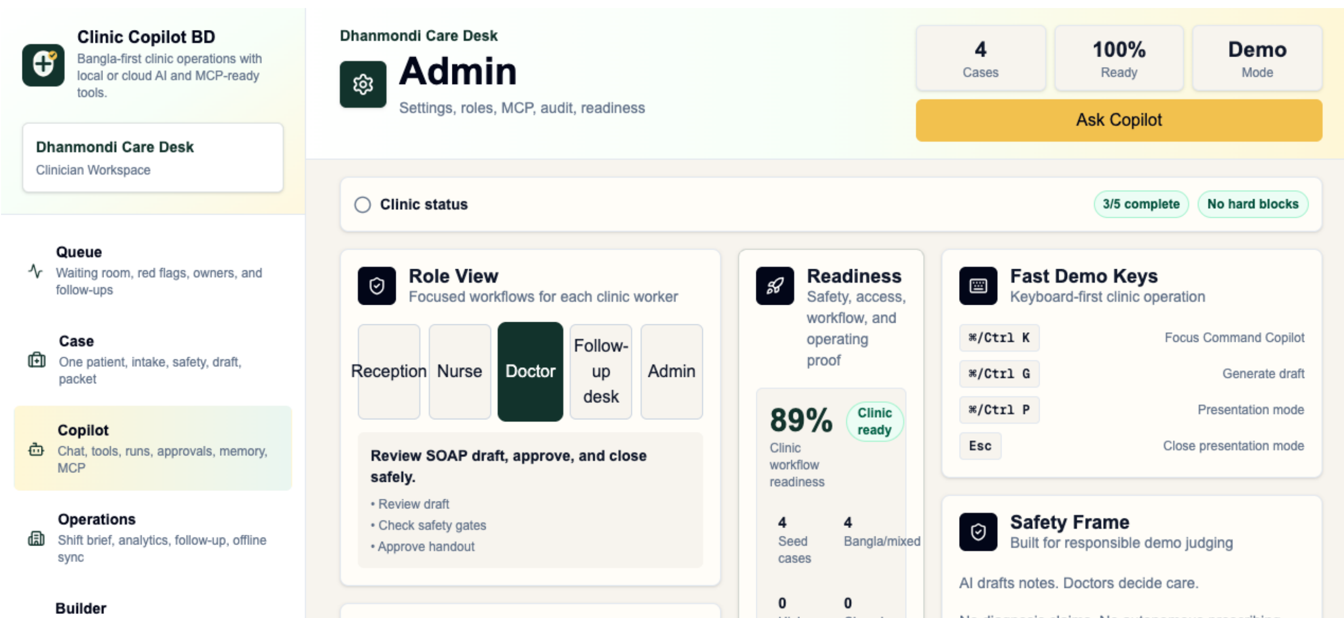


Figure 10: Admin readiness and governance
Admin surfaces support readiness, audit, and integration review.

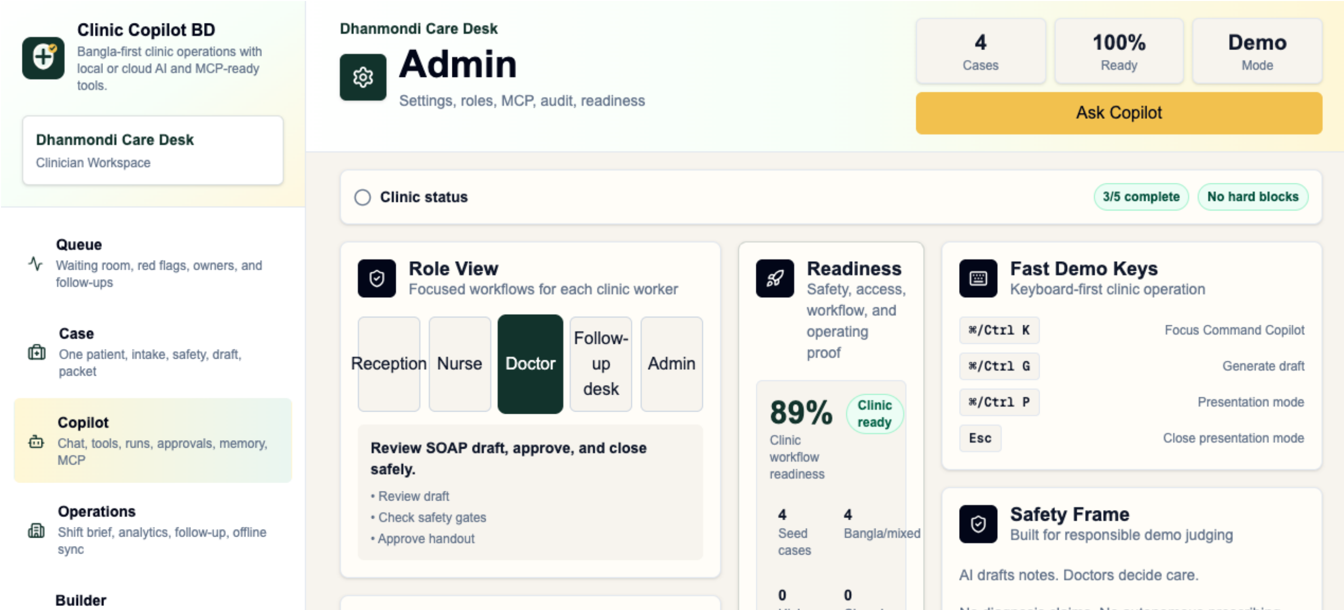


Figure 11: MCP Explorer response

The MCP Explorer proves that external agents can inspect and call bounded clinic workflow tools.

4 Bangladesh Market Context

Bangladesh is a strong proving ground because frontline delivery pressure is real, physician density is low, out-of-pocket burden is high, and community health worker models already exist.

Signal	Data point	Why it matters
Population	173.6 million people in 2024	Small workflow improvements can affect millions of patient encounters.
Physician density	0.722 physicians per 1,000 people in 2023	Frontline workers carry a large share of first-contact care and triage.
Out-of-pocket burden	79.3% of current health expenditure in 2023	Avoidable revisits, confusion, and missed warning signs are expensive for families.
Community clinic model	One community health care provider assisted by two community health workers	The product maps to the team structure used in local clinics.
Private facility footprint	A 2021 brief cites 9,529 DGHS-registered private diagnostic centers in 2020 and reports around 11,940 private hospitals, clinics, and diagnostic centers	Small private facilities are a large fragmented market for lightweight workflow software.
Digital strategy alignment	Bangladesh Digital Health Strategy 2023-2027 emphasizes mHealth, AI, interoperability, privacy, and service innovation	The project fits national direction, not just hackathon novelty.

5 Primary Market: Smaller Hospitals and Overloaded Clinics

The highest-probability first buyer is not a highly resourced tertiary hospital. It is the smaller hospital, local private clinic, diagnostic-linked clinic, NGO clinic, community clinic, or busy outpatient desk where staff are stretched and the workflow is still paper-heavy.

5.1 Why smaller facilities feel the pain first

- One doctor may cover too many patients.
- Nurses often handle intake, triage, patient explanation, and coordination at the same time.
- Reception staff may capture symptoms without a clinical checklist.
- Paper notes, WhatsApp follow-up, printed slips, and verbal handoff are fragmented.
- Patients may return because instructions were unclear.
- High-risk cases can wait in the same queue as routine cases.
- Staff turnover makes consistent protocols hard to maintain.

5.2 How Clinic Copilot BD helps

Operational pain	Product response
Too many patients for the available doctor	Queue risk summary, red-flag lane, missing-vitals checks, concise doctor handoff.
Reception captures incomplete information	Bangla/English intake cleanup, missing question finder, safety gate prompts.
Nurses are overloaded	Staff task list, vitals/allergy blockers, approval readiness, visit closeout checklist.
Patient education takes time	Simple Bangla handout, teach-back checklist, audio script, pictogram plan.
Follow-up is inconsistent	Follow-up due panel, call sheet, reply triage, owner badges.
Documentation slows the visit	Draft SOAP support, referral packet, print workflow, AI run receipt.
Clinic owner needs quality visibility	Audit log, operations pulse, trends, MCP resource access.

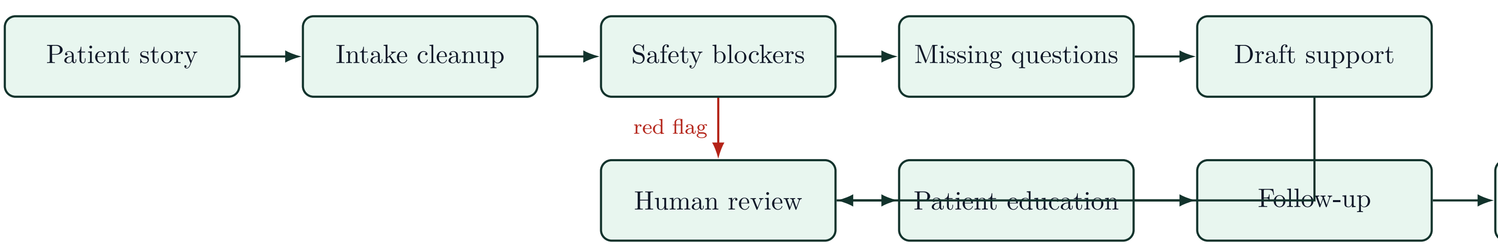
6 Larger Hospital Benefit

Large hospitals have more staff, but they still face bottlenecks in OPD, emergency intake, nursing handoff, patient education, and quality review.

Area	Benefit
Outpatient department	Faster intake normalization, risk sorting, and doctor-ready summaries.
Emergency or urgent desk	Conservative red-flag prompts and escalation blockers before patients wait too long.
Nursing stations	Shared task list, handoff, visit closeout, and approval readiness.
Discharge and patient education	Simple Bangla instructions, teach-back, return warnings, and follow-up ownership.
Training	Synthetic scenarios, scorecards, role-based practice, and reviewable AI run receipts.
Quality and compliance	Audit trail, safety envelope, approval logs, and MCP-readable workflow outputs.
Research and evaluation	Structured case scenarios, measurable outputs, and reproducible MCP tool calls.

7 Workflow Design

7.1 Field-to-clinic loop



7.2 Multi-role workflow

Viewpoint	Main need	Product surface
Reception	Capture story quickly	Intake, queue, command copilot.
Community health worker	Ask what matters next	Safety blockers, missing questions, patient education.
Nurse	Confirm safety gates	Vitals/allergy/return-warning checks, handoff.
Doctor	Review concise support	SOAP support, risk explanation, approval readiness.
Follow-up desk	Close the loop	Scheduler, reply triage, due panel.
Admin/program manager	See service delivery	Trends, audit logs, queue pressure, MCP resources.
External agent	Integrate safely	MCP stdio server and HTTP demo endpoint.

8 AI Safety Architecture

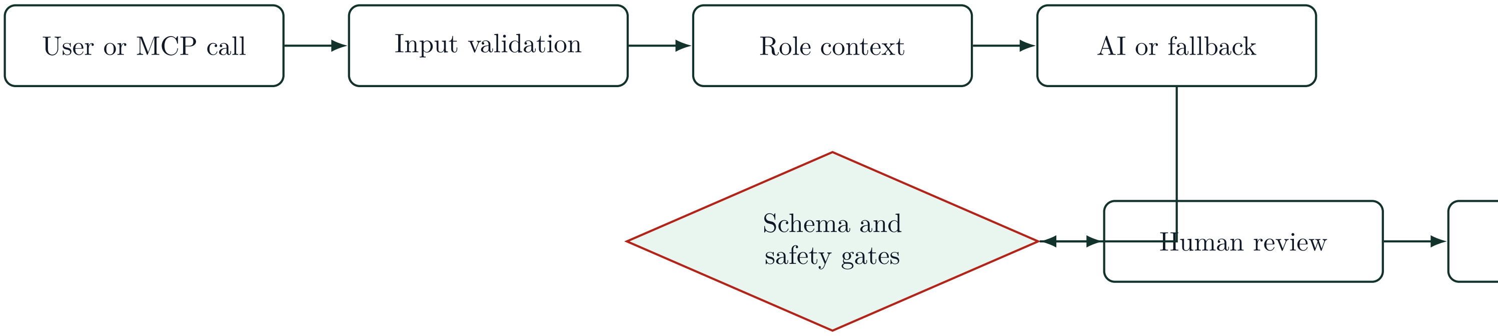
8.1 Allowed AI behavior

- Summarize intake.
- Suggest missing questions.
- Draft handoff notes.
- Draft patient education.
- Flag safety blockers.
- Prepare review packets.
- Explain risk uncertainty.
- Suggest operational next steps.

8.2 Disallowed AI behavior

- Diagnose.
- Prescribe.
- Approve final care.
- Dismiss urgent symptoms.
- Finalize patient-facing instructions without human review.
- Replace clinicians.

8.3 Safety control flow



9 Security Model

Clinic Copilot BD is currently a hackathon prototype using synthetic data and temporary demo authentication. The production security model is designed around patient privacy, role boundaries, model governance, and auditability.

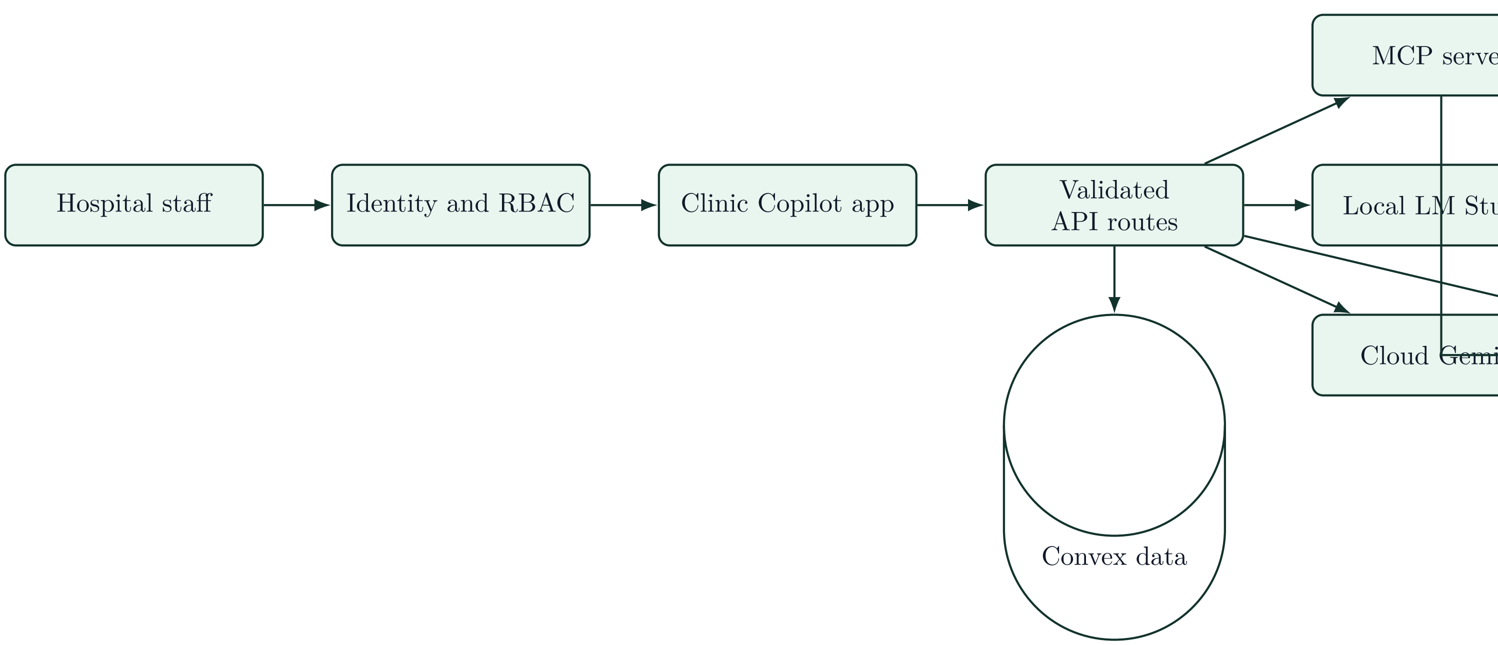
9.1 Current prototype boundaries

- Synthetic demo data only.
- Temporary demo login for testing.
- AI outputs are draft support, not final clinical authority.
- MCP tools return safety envelopes.
- API routes validate inputs and fall back to safe deterministic responses.
- No production PHI storage claim is made in the demo.

9.2 Production hardening plan

Layer	Production control
Identity	Real organization accounts, SSO/OIDC where available, and role-based access for reception, nurse, clinician, admin, auditor, and researcher.
Authorization	Server-side role checks for every case, tool call, approval, export, and patient-facing output. No client-trusted user IDs.
Patient data	Encryption in transit, encryption at rest, tenant isolation, minimal retention, consent flow, and configurable data residency.
Local AI	LM Studio or local OpenAI-compatible models can run inside the clinic/hospital network for privacy-sensitive deployments.
Cloud AI	Cloud calls use redaction, minimum necessary context, provider allowlist, timeout, schema validation, and audit logging.
MCP	Tool allowlist, server-side validation, role-scoped outputs, safety envelopes, and no MCP bypass of clinic permissions.
Audit	Immutable event trail for intake changes, AI calls, approvals, print packets, exports, and MCP calls.
Human review	Patient-facing instructions, clinical handoffs, referrals, and signoff packets remain drafts until a responsible staff member approves.
Research mode	De-identified exports, synthetic scenarios, scenario scorecards, and reproducible tool calls before real-world studies.
Operations	Rate limits, monitoring, error reporting, backup/restore, incident response, and admin controls for disabling model providers or tools.

9.3 Security architecture



10 MCP Strategy

MCP turns Clinic Copilot BD from a standalone app into an agent-ready health workflow layer.

10.1 What MCP enables

- LM Studio can call clinic tools from local models.
- Claude, Codex, Cursor-style clients, and other MCP hosts can spawn the stdio MCP server.
- Public demos can call `/api/mcp` over JSON-RPC.
- External agents can inspect tool schemas before acting.
- Safety rules travel with tool outputs.

10.2 MCP tools

```
clinic.demo_manifest
clinic.list_demo_scenarios
clinic.workflow_brief
clinic.tools.list
clinic.tools.describe
clinic.queue.snapshot
clinic.safety.get_blockers
clinic.print.prepare_packet
clinic.literacy.prepare
clinic.sync.preview_queue
clinic.approval.request
clinic.demo.score_scenario
```

10.3 MCP resources

```
clinic://demo/scenarios
clinic://demo/capabilities
clinic://agents/tool-registry
clinic://safety/gates
```

11 Local and Cloud Model Strategy

11.1 Why local models matter

Field settings may have unstable internet, privacy concerns, low budget, latency constraints, limited cloud availability, and model vendor uncertainty. Clinic Copilot BD supports LM Studio locally through the AI SDK OpenAI-compatible provider.

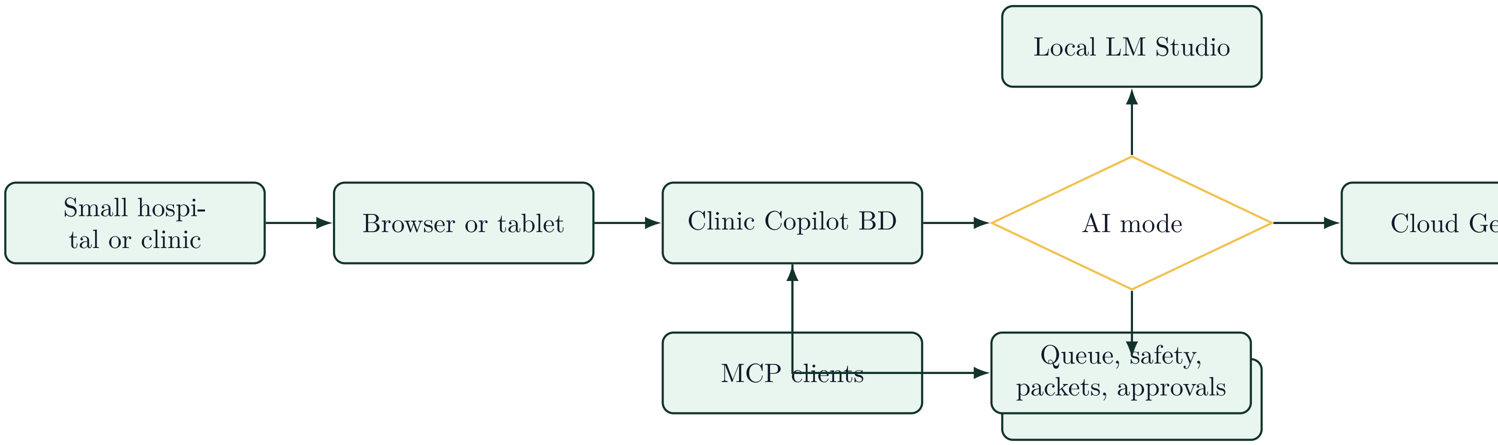
11.2 Why cloud models matter

Cloud models can provide stronger structured output, better multilingual reasoning, easier scaling during pilots, and centralized model updates. Clinic Copilot BD supports Google Gemini through `@ai-sdk/google`.

11.3 Why fallback matters

For demonstrations, training, and offline-style use, the app can return safe deterministic output when no provider is configured. This makes the product testable without fragile API dependencies.

11.4 Deployment pattern



12 Feature Inventory

12.1 AI workflow

Local LM Studio provider; cloud Google Gemini provider; no-key deterministic fallback; mixed Bangla/English intake cleanup; clinical draft generation; SOAP-style support; missing question finder; current-case AI Q&A; command copilot; draft edit support; document extraction; clinic briefing; next-step navigator.

12.2 Safety

Red-flag summary; vitals requirement checks; allergy gap checks; medicine clarity checker; pregnancy/child/chest-pain escalation patterns; approval readiness guard; visit closeout checklist; risk explanation; human-review boundary; AI run receipts; approvals inbox; audit log viewer.

12.3 Health worker operations

Queue board; waiting-time labels; red-flag lane; staff owner badges; follow-up due panel; operations pulse; shift copilot; staff handoff; trend dashboard; low-connectivity review queue; role-aware workspaces.

12.4 Patient communication

Bangla/English handout; printable handout; medicine slip; referral packet; doctor summary; follow-up call sheet; simple Bangla mode; pictogram support; audio script support; teach-back checklist; patient question answer assistant; reply triage; follow-up composer and scheduler.

12.5 Agentic and MCP layer

Stdio MCP server; HTTP JSON-RPC demo endpoint; MCP tools/list and resources/read; `mcp.json` for LM Studio and local MCP hosts; MCP Explorer in Admin; tool registry resource; safety gate resource; demo scenario resource; capability map resource; scenario scorecard.

13 Stack, Scalability, and Modularity

Layer	Technology
Frontend	Next.js 16 App Router, React 19, TypeScript.
Styling	Tailwind CSS 4, responsive mobile-first UI.
Backend	Convex realtime data, Next.js API routes.
AI SDK	Vercel AI SDK 6.
Cloud AI	`@ai-sdk/google`.
Local AI	`@ai-sdk/openai-compatible` for LM Studio.
MCP	`@modelcontextprotocol/sdk`, stdio server, JSON-RPC HTTP demo route.
Validation	Zod schemas.
QA	Bun tests, Biome, Next build, browser QA, MCP smoke.
Runtime	Bun.

13.1 Scalable rollout path

1. Demo mode with synthetic cases.
2. Small hospital or clinic pilot with no real patient storage.
3. Community clinic or NGO pilot with supervised health workers.
4. Local LM Studio deployment for privacy-sensitive training.
5. Cloud provider deployment for stronger model quality.
6. Convex-backed realtime case workflow.
7. MCP integration with health system tools and NGO dashboards.

14 Why Investors Should Care

1. **The pain is operational, frequent, and expensive.** Every clinic day creates repeated intake, triage, documentation, education, follow-up, and handoff work.
2. **Smaller hospitals are a practical wedge.** They need immediate workload relief without buying a heavy hospital information system.
3. **Bangladesh is a high-leverage proving ground.** Large population, low physician density, strong CHW history, high household spending burden, and national digital health priorities create a strong reason to test here.
4. **The product avoids the riskiest AI-health trap.** It does not sell autonomous diagnosis. It sells decision support, documentation acceleration, safety checks, and human-reviewed patient communication.
5. **The architecture is model-flexible.** Local and cloud AI support reduces vendor lock-in and supports low-resource deployment.
6. **MCP makes it platform-shaped.** The app can become a safe tool layer for other agents, not just a standalone UI.

14.1 Investor wedge

Small clinics and hospitals → Daily workload relief → Sticky workflow data → Department expansion → Large hospital and NGO pilots

15 Demo Script for Judges

1. **Field problem.** Show a mixed Bangla-English symptom story and explain that the product asks what is missing, risky, escalated, and understandable to the family.
2. **Intake to structured support.** Load the case workspace and generate draft support.
3. **Safety blockers.** Show missing vitals, allergy checks, medicine safety, and approval readiness.
4. **Patient communication.** Show handout, teach-back, pictogram/audio script support, and follow-up.
5. **Operations.** Show queue pressure, red-flag lane, owner badges, and follow-up due panel.
6. **MCP and model flexibility.** Run MCP smoke or MCP Explorer and explain local LM Studio, cloud Gemini, and fallback operation.

16 Validation Proof

- The refreshed screenshot run completed from a production build through public pages, authenticated workspace, AI actions, drawer views, and MCP Explorer.
- The app is currently reachable at `http://localhost:3000`.
- The screenshot command used a temporary server on port `3003` and saved focused images into `artifacts/health-wo`
- PDF compilation uses `tectonic`.

17 Risks and Honest Limitations

Risk	Current mitigation
Local models may return invalid structured JSON	Timeout, no-retry setting, provider failure fallback, schema tests.
AI could be mistaken for medical authority	Product copy, prompts, safety envelopes, human-review flags, approval inbox.
Connectivity may be weak	Demo fallback and low-connectivity review queue pattern.
Real deployment needs privacy/security hardening	Current demo uses synthetic data and temporary auth only.
Field UX must be tested with real CHWs and clinic staff	Current app is a prototype; next step is supervised usability testing.

18 Next Milestones

1. Add explicit provider status in the UI: local live, cloud live, fallback, timeout.
2. Build offline-first Android wrapper or PWA field mode.
3. Add maternal/child/NCD-specific field protocols.

4. Add supervisor dashboard for CHW and small-hospital programs.
5. Add exportable program reports.
6. Add privacy hardening and production authentication.
7. Pilot with synthetic or anonymized workflows before any real patient use.

19 Source Notes

- World Bank Open Data, Bangladesh population: 173,562,364 in 2024. https://data.worldbank.org/indicator/SP.POP.TOTL?locations=BD
- World Bank Open Data, physicians per 1,000 people: 0.722 in 2023. https://data.worldbank.org/indicator/SH.MED.PHYS.ZS?locations=BD
- World Bank Open Data, out-of-pocket expenditure share of current health expenditure: 79.30711365% in 2023. https://data.worldbank.org/indicator/SH.XPD.OOPC.CH.ZS?locations=BD
- WHO feature story on Bangladesh community health workers and community clinics. https://www.who.int/news-room/feature-stories/detail/bangladesh-community-health-workers-at-the-heart-of-a-stro
- CHW Central profile of BRAC Shasthya Shebika and Shasthya Kormi programs. https://chwcentral.org/the-brac-shasthya-shebika-and-shasthya-kormi-community-health-workers-in-bangladesh/
- Frontline Health Workers Coalition article citing Shasthya Shebika scale and maternal health reach. https://www.frontlinehealthworkers.org/blog/weve-made-staggering-progress-maternal-health-bangl
- Data for Impact brief on private health care data collection in Bangladesh. https://www.data4impactproject.org/wp-content/uploads/2021/05/Improving-Private-Health-Care-Data-Collection_fs-21-517_d4i-1.pdf
- Human Resources for Health article on Bangladesh health workforce crisis. https://pmc.ncbi.nlm.nih.gov/articles/PMC3037300/
- PLOS ONE study on working conditions of clinical health workers in Bangladesh primary and secondary healthcare facilities. https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0294224
- Bangladesh Digital Health Strategy 2023-2027 summary. https://dig.watch/resource/the-bangladesh-digital
- Digital health intervention study in Bangladesh using community health workers, smart health kit, AI-based mobile app, household visits, risk assessment, and digital referrals. https://link.springer.com/article/10.1186/s12889-025-22770-9